

Beyond Carrying Capacity: An Agent-Based Model for the Char Dham Pilgrimage Corridor

1 Background

The Char Dham¹ is one of the world’s largest seasonal pilgrimages, attracting millions of pilgrims within an operating season of roughly six months, with peak-day arrivals exceeding 100,000 (Meru and Pandey, 2024; Kuniyal et al., 2025). This demand is accommodated by a sparse network of narrow mountain roads, single, intermediate, and two-lane(s) for both directions, with limited overtaking opportunities and frequent weather and landslide-induced closures (Chouhan et al., 2023), so that recurrent congestion is an inherent feature of the corridor rather than an occasional disruption. To address some of the challenges on such terrain, the central Government has initiated several infrastructure projects, yet there is no transport model available to assess the impact/ feasibility of these projects. Additionally, most simulation frameworks assume the same link dynamics, which may not be realistic in practice. For instance, passing/ overtaking is unlikely on some of the roads, whereas it is possible on the rest of the network. Naturally, a transport simulation model may be a cornerstone for demonstrating sufficient empirical credibility, allowing planners to investigate infrastructure and demand-management interventions before implementation, which is especially valuable in an ecologically fragile Himalayan setting. The present study addresses this gap with an open-source, agent-based simulation framework for the Char Dham network, built in MATSim, that combines a calibrated mountain-road network with a synthetic pilgrim population derived from official registration data and is validated against independent travel-time and observed pilgrim footfall.

2 Methodology

Network: The road network of the study region was extracted from OpenStreetMap and verified for key corridors using field-observed data. The first novel contribution is allowing different links with different link dynamics, for which single-lane mountain segments were identified for FIFO (First-In-First-Out) link dynamics representing the physical constraints of overtaking, while all other links retained standard passing link dynamics, consistent with prior MATSim applications to heterogeneous traffic (Agarwal et al., 2015, 2017). A large number of sub-100 m links arising from OSM digitization were contracted to reduce computational cost without altering connectivity.

Demand: Travel demand was derived from the 2024 Char Dham registration database (2,518,464 valid pilgrim records after cleaning), reshaped into 6,033,400 dated shrine-visit events and chronologically ordered into observed itinerary chains. The dominant pattern was the classical west-to-east (see Figure 1) sequence Yamunotri → Gangotri → Kedarnath → Badrinath (27.6% of itineraries), with transition probabilities as high as 97.1% (Kedarnath → Badrinath). An 8-day PEAK demand window (717,335 represented passengers, 9-16 June 2024) was selected to capture the observed route-duration distribution and weekday/weekend heterogeneity. Registration-derived mode shares (bus/minibus, taxi, car, motorbike) and mode-specific occupancy ranges were used to convert passenger demand into 46,538 vehicle/group plans (down-sampled to 23,269 at a 0.5 sampling fraction), with each shrine visit expanded into a realistic multi-stop route through intermediate towns and overnight halts.

¹*Dham* is a Sanskrit term meaning a sacred abode or pilgrimage site; in this context, it refers to the four sacred Himalayan shrines of Yamunotri, Gangotri, Kedarnath, and Badrinath in Uttarakhand, India.

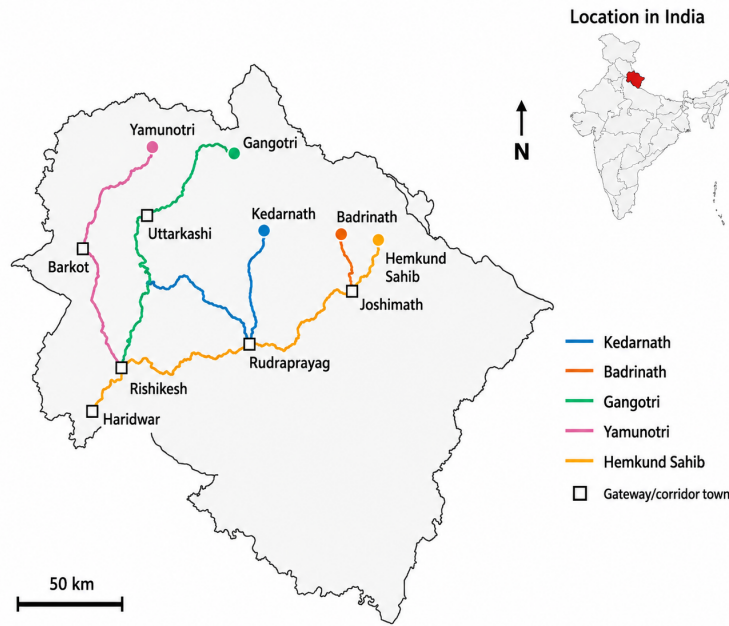


Figure 1: Study Area

Simulation and calibration. Each scenario was run for 100 iterations using MATSim’s standard co-evolutionary process of plan execution, scoring, and replanning. Agent utility used the default MATSim scoring function, with mode- and activity-specific utility parameters. Replanning combined the default route and time-mutation strategies with a custom overnight/ intermediate location-choice. The latter is integrated to avoid or reduce nighttime trips under the current practice. The module lets agents search among alternative facilities within a defined radius (e.g., 30 km) of their currently selected activity. In order to introduce different link dynamics, first, the desired link dynamics are set in link attributes, and then, `qsimConfigGroup` is assigned for each link based on the set attribute (i.e., `FIFO` or `PassingQ`), which triggers respective `QSim` in the mobility simulation.

3 Results and Validation

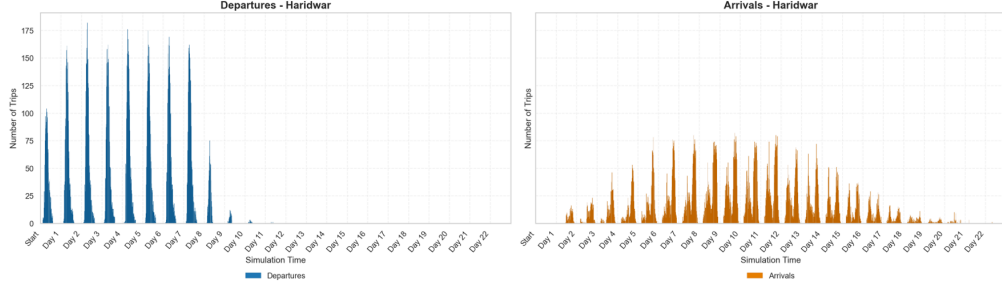
3.1 Results

Arrival and Departure Patterns The temporal arrival and departure patterns show how the loaded demand propagates through the multi-day pilgrimage simulation. Figure 2 presents Haridwar, the primary origin and return location for the simulated pilgrimage trips, together with Kedarnath as a representative shrine. At Haridwar, departures are concentrated during the initial demand-loading period, whereas arrivals are distributed over the subsequent days as pilgrims complete their itineraries and return to the gateway. In contrast, arrivals and departures at Kedarnath occur with a temporal lag and are distributed across the simulation period, reflecting the time required for pilgrims to progress through their respective multi-day trip chains. This illustrates the temporal propagation of the initially loaded demand through the pilgrimage circuit.

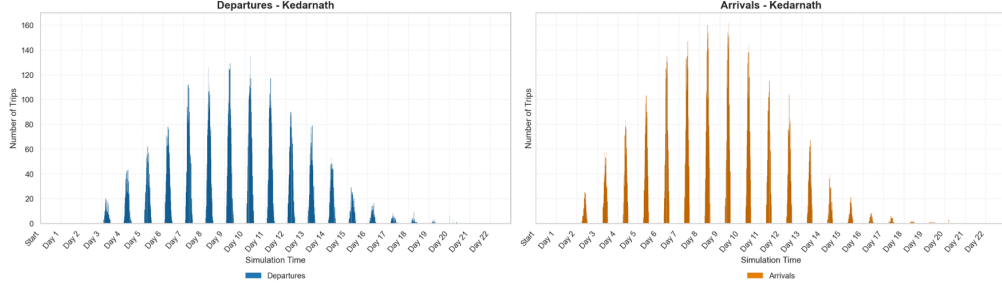
3.2 Validation

The simulation framework was evaluated through three validation exercises: shrine-level pilgrim counts, trip-chain durations, and corridor travel times. Together, these assessments examine whether the model reproduces the observed magnitude of shrine arrivals, the temporal plausibility of multi-day pilgrimage itineraries, and the travel conditions experienced along the road network.

Shrine-level count validation. Figure 3a compares the simulated pilgrim counts with the corresponding 2024 field observations during the active-arrival period at each shrine. To reduce the



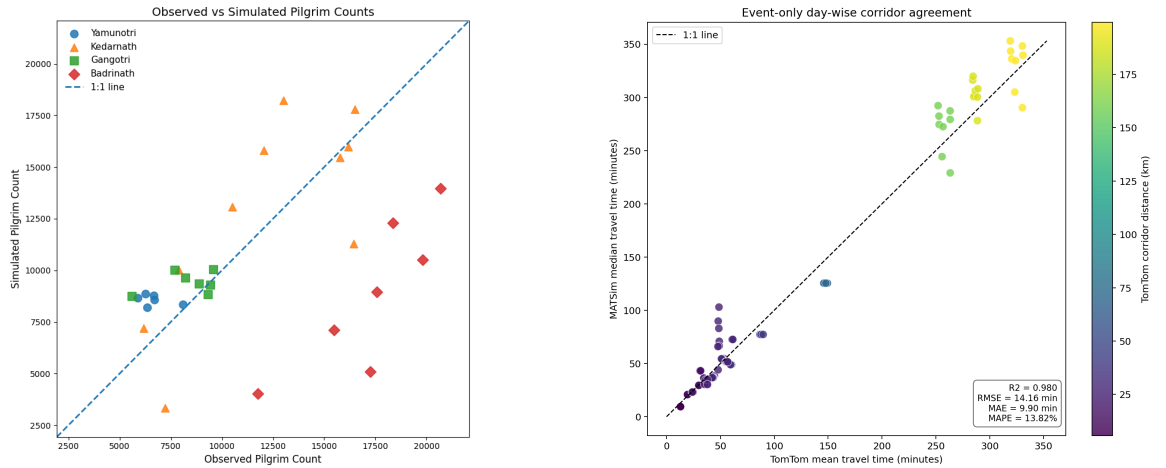
(a) Departures and arrivals at Haridwar.



(b) Departures and arrivals at Kedarnath.

Figure 2: Temporal propagation of simulated pilgrimage demand, illustrated through departure and arrival patterns at (a) Haridwar, the initial and return gateway, and (b) Kedarnath, a representative shrine along the pilgrimage itinerary.

influence of simulation start-up and end-of-horizon effects, the validation was restricted to selected simulation days: D4-D9 for Yamunotri, D6-D15 for Kedarnath, D4-D10 for Gangotri, and D3 and D5-D10 for Badrinath, resulting in 30 shrine-day observations. The simulated pilgrim counts were estimated from the simulated vehicle groups using the same vehicle-occupancy assumptions adopted during plan generation and were then compared with the observed pilgrim visit counts. Yamunotri, Gangotri, and Kedarnath simulation counts are closely aligned with the observed pilgrim counts, whereas Badrinath simulation counts are underestimates. One possible source of these shrine-specific differences is the assumption of a fixed modal composition across all shrines, whereas the actual mode share likely varies by shrine.



(a) Observed 2024 pilgrim counts versus simulated shrine-level counts. (b) Daily corridor travel-time validation comparing simulated and TomTom-observed travel times.

Figure 3: Validation of simulated pilgrimage demand and corridor travel times.

Corridor travel-time validation. Figure 3b illustrates the daily comparison between simulated and TomTom-observed travel times along the selected pilgrimage corridors. The simulated travel times show strong agreement with the observations across multiple temporal resolutions. The coefficient of determination was $R^2 = 0.989$ at the aggregate level, $R^2 = 0.980$ at the daily level, and $R^2 = 0.959$ at the hourly level.

Trip-chain and duration validation. The temporal realism of the simulated pilgrimage itineraries was further examined through the distribution and completion times of the dominant trip chains. The most common pilgrimage chains among the 23,269 simulated travel groups, which together account for 82.4% of all simulated groups. The complete Yamunotri-Gangotri-Kedarnath-Badrinath circuit is the most frequent itinerary, representing 29.0% of all groups and requiring an average of 9 days to complete. The Kedarnath-Badrinath chain accounts for 22.3% of groups and has an average duration of 6 days. Kedarnath-only and Badrinath-only trips represent 20.2% and 15.6% of groups, with average completion times of 5 and 2 days, respectively. Overall, the simulated trip duration increases with the number of shrines visited and the spatial extent of the itinerary. The resulting duration patterns are consistent with the corresponding ground-truth trip-duration characteristics, supporting the temporal plausibility of the simulated multi-day pilgrimage chains.

4 Conclusion

This study develops and validates an itinerary-aware, agent-based simulation framework for the Char Dham pilgrimage system. By combining registration-derived multi-day travel demand with a heterogeneous mountain-road network, the model captures the key spatial and temporal characteristics of pilgrimage mobility. The validation results indicate that the framework reproduces overall travel patterns and corridor conditions with reasonable accuracy, while also highlighting areas that require further calibration at the shrine level. The model therefore provides a useful basis for evaluating future transport-management and infrastructure strategies in the Char Dham region and can be adapted to other capacity-constrained pilgrimage and mountain transport systems.

Presentation mode: The author intends to present this work remotely and kindly requests consideration for the remote presentation option.

References

- A. Agarwal, M. Zilske, K. R. Rao, and K. Nagel. An elegant and computationally efficient approach for heterogeneous traffic modelling using agent based simulation. *Procedia Computer Science*, 52 (C):962–967, 2015. ISSN 1877-0509. doi: 10.1016/j.procs.2015.05.173.
- Amit Agarwal, Gregor Lämmel, and Kai Nagel. Incorporating within link dynamics in an agent-based computationally faster and scalable queue model. *Transportmetrica A: Transport Science*, 14(5-6):520–541, 2017. ISSN 2324-9943. doi: 10.1080/23249935.2017.1364802.
- Shivani Chouhan, Annegret H. Thieken, Philip Bubeck, and Mahua Mukherjee. Role of tourism on disaster recovery: A case study of uttarakhand, india. *International Journal of Disaster Risk Reduction*, 95:103878, 2023. ISSN 2212-4209. doi: 10.1016/j.ijdrr.2023.103878.
- Jagdish Chandra Kuniyal, Priyanka Maiti, Nidhi Kanwar, Rupesh Dhyani, and Mahesha Nand. Carrying capacity and strategic planning for sustainable tourism practices in the char dham from the western himalaya, india. *Scientific Reports*, 15(1), 2025. ISSN 2045-2322. doi: 10.1038/s41598-025-20166-8.
- Lungyina B. Meru and Rajiv Pandey. Climate change ecological vulnerability and hotspot analysis of himalayan forests in north-eastern region, india. *Environmental and Sustainability Indicators*, 24:100472, 2024. ISSN 2665-9727. doi: 10.1016/j.indic.2024.100472.